1. algorithm_name (inputs)
2. -- inputs : P
3. -- outputs: Q

This specification means that if the algorithm starts with inputs satisfying P, then it will finish with the outputs satisfying Q.

A double dash -- indicates that the rest of the line is a comment. Comments are statements which are used to annotate a program for the human readers and not executed by the computer. Comments at crucial points of flow are useful and even necessary to understand the algorithm. In our algorithmic notation, we use double dashes (--) to start a comment line. (In C++, a double slash // indicates that the rest of the line is a comment).

Example 6.6. Write the specification of an algorithm to compute the quotient and remainder after dividing an integer A by another integer B. For example,

divide (22, 5) = 4, 2

divide (15, 3) = 5 , 0

Let A and B be the input variables. We will store the quotient in a variable q and the remainder in a variable r. So q and r are the output variables.

What are the properties of the inputs A and B?

1. A should be an integer. Remainder is meaningful only for integer division and
2. B should not be 0, since division by 0 is not allowed.

We will specify the properties of the inputs as

-- inputs: A is an integer and B ≠ 0

What is the desired relation between the inputs A and B and the outputs q and r?

1. The two outputs q (quotient) and r (remainder) should satisfy the property

   A = q X B + r, and
2. The remainder r should be less than the divisor B,

   0 ≤ r < B

Combining these requirements, we will specify the desired input-output relation as

-- outputs: A = q X B + r and 0 < r < B.

The comment that starts with -- inputs: actually is the property of the given inputs. The comment that starts with -- outputs: is the desired relation between the inputs and the outputs. The specification of the algorithm is

1. **divide (A , B)**
2. **-- inputs: A is an integer and B ≠ 0**
3. **-- outputs : A = q X B + r and 0 ≤ r < B**

Specification format: We can write the specification in a standard three part format:

- The name of the algorithm and the inputs.
- Input: the property of the inputs.
- Output: the desired input-output relation.

The first part is the name of the algorithm and the inputs. The second part is the property of the inputs. It is written as a comment which starts with -- inputs: The third part is the desired input-output relation. It is written as a comment which starts with -- outputs:. The input and output can be written using English and mathematical notation.

Example 6.7. Write the specification of an algorithm for computing the square root of a number.

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